

A CFG is a set of rules/productions used to generate patterns of strings.

CFG $\rightarrow$ 4 tuples $(\underbrace{V}_N, \underbrace{T}_\Sigma, \underbrace{P}_R, S)$

$V \Rightarrow$ Variable / Non terminals set

$T =$ set of terminals

$P =$ productions / Rules

$S =$ start variable

CFG for $\{0^n 1^n \mid n \geq 1\}$

$0^1 1^1 = \{01\}$

$0^2 1^2 = \{0011\}$

$S \Rightarrow$

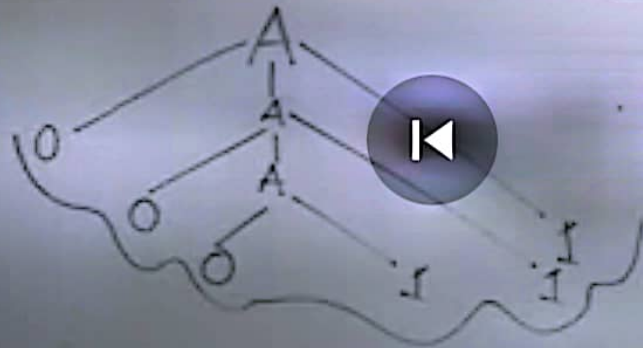
Context-free Grammar (CFG)

* A CFG is a set of rules/productions used to generate patterns of strings.

CFG
pattern $\rightarrow 000111$

$$A \rightarrow 01 \mid 0A1 \quad \text{or, } A \rightarrow 01$$

$$A \rightarrow 0A1$$



$$L = \{ \text{strings containing 0 and 1} \}$$

$$= \{ 01, 001, 011, 0011, 0111, 000111, \dots \}$$

$$L = \{ 0^n 1^n \mid n \geq 1 \}$$

or

$$01$$

$$0011$$

$$000111$$

$n=1; 0^1 1^1 = 01$
 $n=2; 0^2 1^2 = 0011$
 $n=3; 0^3 1^3 = 000111$

CFG

Example-1

$$L = \{a^n \mid n \geq 0\}$$

$$= \{\epsilon, a, aa, \dots\}$$

CFG

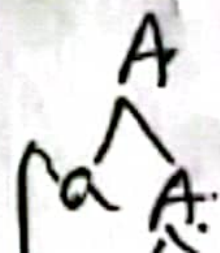
$$A \Rightarrow aA \mid \epsilon$$

$$A \Rightarrow \epsilon$$

$$A \Rightarrow aA$$

$$\Rightarrow aaA$$

$$\Rightarrow aa\epsilon$$



$$a^*$$

$$b^*$$

$$A \Rightarrow bA \mid \epsilon$$

$$c^*$$

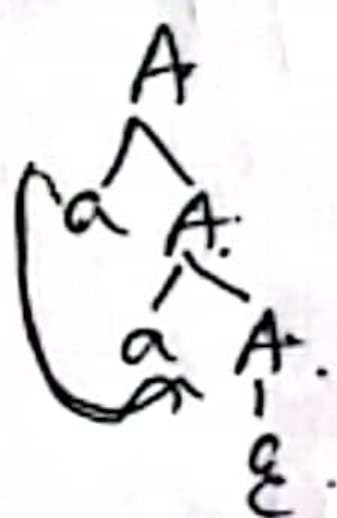
$$A \Rightarrow cA \mid \epsilon$$

Example-2

$$L = \{a^n \mid n \geq 1\}$$

$$= \{a, aa, aaa, \dots\}$$

$$A \Rightarrow aA \mid a$$



Example - 3

$L = \{\text{set of all strings over } a, b\}$

$(a+b)^*$

~~$(a|b)^*$~~

~~a^*~~

$A \rightarrow aA \mid bA \mid \epsilon$

$A \rightarrow aA \mid \epsilon$

~~$(a+b)^*$~~

$A \rightarrow aA \mid bA \mid \epsilon$

Example - 4

CFG for set of all strings which length at least 2

$$L = \{ \underline{aa}, \underline{ab}, \underline{ba}, \underline{bb}, \underline{aaa}, \dots \}$$

$$\begin{array}{ccc} \underline{(a+b)} & \underline{(a+b)} & \underline{(a+b)^*} \\ A & A & B \end{array}$$

$$S \rightarrow AAB$$

$$A \rightarrow a|b$$

$$B \rightarrow aB | bB | \epsilon$$

$$\begin{array}{l} a^* \\ A \rightarrow aA | \epsilon \\ (a+b)^* \\ A \rightarrow aA | bA | \epsilon \end{array}$$

Example - 5

CFG for set of all strings ~~which length at most 2~~
in ~~which~~ ^ε of at least 3 0's

$$\textcircled{E} \cdot \overbrace{EOEOEOE}^{(0+1)^*}$$

$$S \rightarrow EOEOEOE$$
$$E \rightarrow 0E \mid 1E \mid \epsilon$$

$$\boxed{1.0.00}$$

$$S \Rightarrow EOEOEOE$$
$$\Rightarrow 1EOEOEOE$$
$$\Rightarrow 10EOEOE$$
$$\Rightarrow 100EOE$$
$$\Rightarrow 1000E$$
$$\Rightarrow 1000$$

Example - 6

set of all strings which length at most 2

$$L = \{ \epsilon, a, b, aa, ab, ba, bb \}$$

$$RE \rightarrow \underbrace{(a+ b+ \epsilon)}_A \cdot \underbrace{(a+ b+ \epsilon)}_A$$

$$S \rightarrow AA$$

$$A \rightarrow a | b | \epsilon$$

Example - 7

with b

Example-7

starts with a and ends with b

$$a \underbrace{(a+b)^*}_A b$$

$$\begin{aligned} S &\rightarrow aAb \\ A &\rightarrow aA \mid bA \mid \epsilon \end{aligned}$$

a b a b



Example-8

starts and ends with different symbol

$$a \underbrace{(a+b)^*}_A b \quad | \quad b \underbrace{(a+b)^*}_A a$$

$$\checkmark a(a+b)^*b + b(a+b)^*a$$

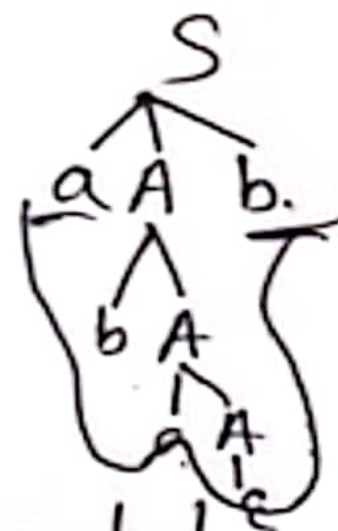
Starts with a

$$a \underline{(a+b)^*} b$$

A

$$\begin{aligned} S &\rightarrow aAb \\ A &\rightarrow aA \mid bA \mid \epsilon \end{aligned}$$

a b a b



abab

Example - 8

Starts and ends with different Symbol

$$a \underline{(a+b)^*} b \quad | \quad b \underline{(a+b)^*} a$$

$$\underline{a \underline{(a+b)^*} b} + \underline{b \underline{(a+b)^*} a}$$

A A

$$S \rightarrow aAb \mid bAa$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

Example - 9

Starts and ends with same symbol

$$\underline{a} \frac{(a+b)^*}{A} \underline{a} \quad | \quad \underline{b} \frac{(a+b)^*}{A} \underline{b} \quad | \underline{\epsilon} | \underline{a} | \underline{b}$$

$$S \rightarrow aAa \mid bAb \mid \epsilon \mid a \mid b$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

Example - 10

Example - 10

Even length string

$((a+b) \dots (a+b))^*$

$S \rightarrow BS | \epsilon$

$B \rightarrow A.A$

$A \rightarrow a | b$

Ambiguity

If a grammar generates the same string in several different ways, we say that the string is derived ambiguously in that grammar. If a grammar generates some string ambiguously, we say that the grammar is ambiguous. Example:

(16)

$S \rightarrow S + S \mid S * S \mid a$ → Ambiguous Grammar

$S \rightarrow S + S$
 $\rightarrow a + S$
 $\rightarrow a + S * S$
 $\rightarrow a + a * S$
 $\rightarrow a + a * a$

$S \rightarrow S * S$
 $\rightarrow S + S * S$
 $\rightarrow a + S * S$
 $\rightarrow a + a * S$
 $\rightarrow a + a * a$

pattern/string → $a + a * a$

↓
Ambiguously generated string.

Remove Null Productions from CFG

① $S \rightarrow ABC$
 $A \rightarrow aA \mid \epsilon$
 $B \rightarrow bB \mid \epsilon$
 $C \rightarrow c$

$A \rightarrow \epsilon$,

$S \rightarrow ABC \mid Bc$
 $A \rightarrow aA \mid a$
 $B \rightarrow bB \mid \epsilon$
 $C \rightarrow c$

Removing $B \rightarrow \epsilon$,

$S \rightarrow ABC \mid Bc \mid Ac \mid C$
 $A \rightarrow aA \mid a$
 $B \rightarrow bB \mid b$
 $C \rightarrow c$

② $S \rightarrow ABAC$
 $A \rightarrow aA \mid \epsilon$
 $B \rightarrow bB \mid \epsilon$
 $C \rightarrow c$

③ $S \rightarrow aS \mid A$
 $A \rightarrow \epsilon$

Remove Null from LR from LR

$$\begin{aligned} \textcircled{2} \quad S &\rightarrow ABAC \\ A &\rightarrow aA \mid \epsilon \\ B &\rightarrow bB \mid \epsilon \\ C &\rightarrow c \end{aligned}$$

Removing $A \rightarrow \epsilon$.

$$S \rightarrow ABAC \mid BAC \mid ABC \mid BC$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid \epsilon$$

$$C \rightarrow c$$

Removing $B \rightarrow \epsilon$.

$$S \rightarrow ABAC \mid BAC \mid ABC \mid BC \mid AC \mid C$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow c$$

$$\begin{aligned} \textcircled{1} \quad S &\rightarrow ABAC \\ A &\rightarrow aA \mid \epsilon \\ B &\rightarrow bB \mid \epsilon \\ C &\rightarrow c \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad S &\rightarrow aS \mid A \\ A &\rightarrow \epsilon \end{aligned}$$

Remove Null Productions from CFG

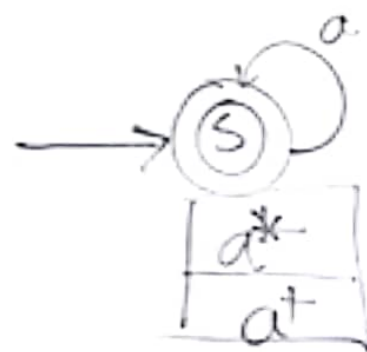
② $S \rightarrow aS \mid A$
 $A \rightarrow \epsilon$

Removing $A \rightarrow \epsilon$,

$$S \rightarrow aS \mid \epsilon$$

Removing $S \rightarrow \epsilon$,

$$S \rightarrow aS \mid \epsilon \mid a$$



$$a \geq 0$$

$$a \geq 0$$

② $S \rightarrow ABAC$
 $A \rightarrow aA \mid \epsilon$
 $B \rightarrow bB \mid \epsilon$
 $C \rightarrow c$

③ $S \rightarrow aS \mid A$
 $A \rightarrow \epsilon$

Remove Unit Productions from CFG

① $A \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow C|b$
 $C \rightarrow D$
 $D \rightarrow E$
 $E \rightarrow a$

Solⁿ:

$A \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow a|b$
 $C \rightarrow a$
 $D \rightarrow a$
 $E \rightarrow a$

~~$B \rightarrow C$~~ $B \rightarrow C \rightarrow D \rightarrow E$
 ~~$C \rightarrow D$~~
 ~~$D \rightarrow E$~~

Remove Unreachable Variables:

$A \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow a|b$

$S \rightarrow aA|B$
 $A \rightarrow ba|bb$
 $B \rightarrow A|bba$

Remove Unit Productions from CFG

$S \rightarrow aA \mid B$
 $A \rightarrow ba \mid bb$
 $B \rightarrow A \mid bba$

n:

$S \rightarrow aA \mid ba \mid bb \mid bba$
 $A \rightarrow ba \mid bb$
 $B \rightarrow ba \mid bb \mid bba$

Remove Unreachable Variables:

$S \rightarrow aA \mid ba \mid bb \mid bba$
 $A \rightarrow ba \mid bb$

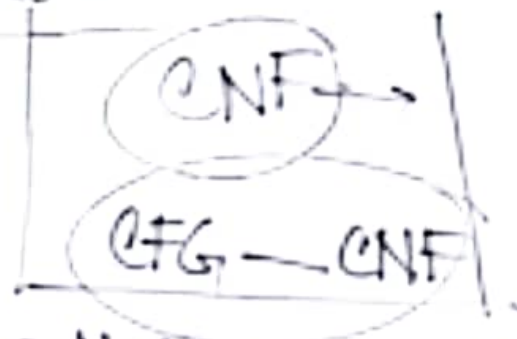


Remove Unit Productions from CFG

Remove Unreachable Variables:

$S \rightarrow aA \mid ba \mid bb \mid bba$

$A \rightarrow ba \mid bb$



Second $\rightarrow$ Null Production Removal .
Third $\rightarrow$ Unit " "

Context Free Grammar - to Chomsky Normal Form

① S →
A →
B →

$A \rightarrow (a)$
 $A \rightarrow (BC)$ ↓ RHS

$a \rightarrow \text{terminal}$
 $BC \in V$

Context Free Grammar -> Chomsky Normal Form

$S \rightarrow ASA \mid aB$
 $A \rightarrow B \mid S$
 $B \rightarrow b \mid \epsilon$

Step 1: Add ($S_0 \rightarrow S$)

$S_0 \rightarrow S$
 $S \rightarrow ASA \mid aB$
 $A \rightarrow B \mid S$
 $B \rightarrow b \mid \epsilon$

Step 2: Remove Null Production:

$B \rightarrow \epsilon, A \rightarrow \epsilon$

Removing $B \rightarrow \epsilon$,

$S_0 \rightarrow S$
 $S \rightarrow ASA \mid aB \mid a$
 $A \rightarrow B \mid S \mid \epsilon$
 $B \rightarrow b$

Removing $A \rightarrow \epsilon$,

$S_0 \rightarrow S$
 $S \rightarrow ASA \mid aB \mid a \mid SA \mid AS \mid S$
 $A \rightarrow B \mid S$
 $B \rightarrow b$

Step 3: Remove Unit Productions:

$S_0 \rightarrow S, S \rightarrow S, A \rightarrow B, A \rightarrow S$

$S_0 \rightarrow ASA \mid aB \mid a \mid SA \mid AS$

$S \rightarrow ASA \mid aB \mid a \mid SA \mid AS$

$A \rightarrow b \mid ASA \mid aB \mid a \mid SA \mid AS$

$B \rightarrow b$

$X \rightarrow SA$

Step 1:

$S_0 \rightarrow \cancel{AX} | \cancel{aB} | a | \cancel{SA} | \cancel{AS}$
 $S \rightarrow \cancel{AX} | \cancel{aB} | a | SA | AS$
 $A \rightarrow \cancel{b} | \cancel{AX} | \cancel{aB} | a | \cancel{SA} | \cancel{AS}$
 $B \rightarrow \cancel{b}$
 $X \rightarrow SA$

Context Free Grammar - to Chomsky Normal Form

Step 2: Remove Null Production:

$B \rightarrow \epsilon, A \rightarrow \epsilon$

Removing $B \rightarrow \epsilon$,

$S_0 \rightarrow S$
 $S \rightarrow ASA | aB | a$
 $A \rightarrow B | S | \epsilon$
 $B \rightarrow b$

Removing $A \rightarrow \epsilon$,

$S_0 \rightarrow S$
 $S \rightarrow ASA | aB | a | SA | AS | S$
 $A \rightarrow B | S$
 $B \rightarrow b$

Step 3: Remove Unit Productions:

$S_0 \rightarrow S, S \rightarrow S, A \rightarrow B, A \rightarrow S$

$S_0 \rightarrow ASA | aB | a | SA | AS$
 $S \rightarrow ASA | aB | a | SA | AS$
 $A \rightarrow b | ASA | aB | a | SA | AS$
 $B \rightarrow b$

CNF

$A \rightarrow a$

$A \rightarrow B$

1.

Context Free Grammar $\rightarrow$ Chomsky Normal Form

Step 5:

$S \rightarrow \textcircled{AX} \mid \textcircled{YB} \mid \textcircled{a} \mid \textcircled{SA} \mid \textcircled{AS}$
 $S \rightarrow \textcircled{AX} \mid \textcircled{YB} \mid \textcircled{a} \mid \textcircled{SA} \mid \textcircled{AS}$
 $A \rightarrow \textcircled{b} \mid \textcircled{AX} \mid \textcircled{YB} \mid \textcircled{a} \mid \textcircled{SA} \mid \textcircled{AS}$
 $B \rightarrow \textcircled{b}$
 $X \rightarrow \textcircled{SA}$
 $Y \rightarrow \textcircled{a}$

Step 3: Remove Unit Productions:

$S \rightarrow S, S \rightarrow S, A \rightarrow B, A \rightarrow S$

$S \rightarrow \textcircled{ASA} \mid \textcircled{AB} \mid \textcircled{a} \mid \textcircled{SA} \mid \textcircled{AS}$

$S \rightarrow \textcircled{ASA} \mid \textcircled{AB} \mid \textcircled{a} \mid \textcircled{SA} \mid \textcircled{AS}$

$A \rightarrow \textcircled{b} \mid \textcircled{ASA} \mid \textcircled{AB} \mid \textcircled{a} \mid \textcircled{SA} \mid \textcircled{AS}$

$B \rightarrow \textcircled{b}$

CNF